



Department of Information Technology
Academic Year 2024–2025 (Odd Semester)

Degree, Semester & Branch : III Semester, B.Tech, IT
Course Code / Title : CD3291/Data Structures and Algorithms
Name of the Faculty member : Mrs. G. Mareeswari

Innovative Practice Description

Unit / Topic: Unit I - Abstract Data Types / Recursion

Course Outcome: CO1

Topic Learning Outcome: TLO4

Activity Chosen: Flipped Classroom

Justification:

The flipped classroom is an instructional strategy where traditional teaching methods are inverted or "flipped." Students are assigned materials to study before the class. This activity allows them to gain exposure to the content independently at their own pace. Classroom time is then dedicated to interactive learning activities, discussions. It assists the instructor/faculty in redefining in-class activities, assigning homework, and maintaining student interest in the concept. The recursion concept was already known by the students during the previous semester so this flipped classroom activity is planned for the topic recursion.

Time Allotted for the Activity: 20 Minutes

Details of the Implementation:

- The learning materials were posted to the students in canvas two days before the topic discussion.
- The students already learned the concept in the previous semester.
- The students are instructed to study the material and take notes on the topic based on their understanding.
- On the day of the activity the students were grouped into a team of 2 members and they are asked to discuss about the topic they learnt in their home through the study materials for the duration of 10 minutes.

- After the discussion one member from each team had a chance to present and share the idea of Recursion to the entire class as shown in Fig.1 &2

Plan: Separate student's teams are formed and made them to explore as a group.

Identify and Share: Related materials are identified and posted relevant materials /notes of recursion topic. Aashika. A, and Parasuram. C. A are requested to present the concepts they have learnt and share it to the entire class students.

Evaluate:

- I have prepared few questions related to the content shared and ask those questions in next class session and make the students to write/ present in class.
- Make the students to find out the answer by their own by learning.

CO–PO/ PSOMapping:

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO1	2	2	2	1	1	1

(1– Low 2 –Moderate 3 – High)

PO/ PSOMapped:

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
	2	2	2	1	1	1
Justification for correlation	Apply basic Knowledge and fundamentals in engineering to understand the recursion concept	Using the concept recursion we can able to analyze and simplify complex Engineering problems.	Able to design and develop solutions for various complex problem.	Students involved in this activity as an individual and also as a team.	Communicate/ share the ideas with other students	Ability to find solution for data science problem using the recursion concept.

Images/ Screenshot of the practice:



Figure 1 and 2 shows the Glimpses of presentation done by Aashika. A and Parasuram. C. A during the flipped class room Activity.

Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

- Student felt good, since they can study at their own pace/time.
- They felt that through this type of active learning, they can explore themselves more.

❖ *Benefit of the practice:*

- This activity provides one-to-one time between teacher and student.
- It represents the way of learning by collaboration within the students.
- Students learn at their own pace.
- Practical things—like missing class due to illness – become less problematic.
- It encourages students to come to class prepared.

❖ *Challenges faced in implementation:*

- The activity depends on student preparation—few students did not come prepared.
- The depth of the subject can be dictated by the student themselves or the group the student is working with.
- The time and effort required from a teacher's perspective initially when creating the flipped class material is higher than for a traditional class.

References:

- ❖ <https://www.teachthought.com/learning/the-definition-of-the-flipped-classroom/#:~:text=A%20flipped%20classroom%20is%20a,the%20students%20independently%20at%20home.>
- ❖ https://en.wikipedia.org/wiki/Flipped_classroom
- ❖ [youtube.com/watch?v=BCIxikOq73Q](https://www.youtube.com/watch?v=BCIxikOq73Q)
- ❖ <https://omerad.msu.edu/teaching/teaching-skills-strategies/27-teaching/162-what-why-and-how-to-implement-a-flipped-classroom-model>

Signature of the Faculty

Signature of the HOD